BD-13 1046: a 5.078-day eclipsing binary

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Summary

BD-13 1046 (TIC 348773952) is a detached eclipsing binary with a period of 5.078277 ± 0.000001 d, not previously catalogued as variable.

- **Eclipses:** TESS shows a primary eclipse 13.0% deep (0.151 mag) and a secondary 3.0% deep (0.033 mag) at phase 0.5001. The primary lasts 2.6 hours.
- **Orbit:** the period agrees with the star's Gaia DR3 spectroscopic orbit (5.07811 ± 0.00012 d, 1.4σ), and the secondary at phase 0.500 is consistent with Gaia's near-circular orbit (e = 0.017 ± 0.007).
- **Ground data:** ASAS-SN V photometry from 2013 to 2018 shows the primary eclipse at the TESS ephemeris, so the period holds over about 13 years.
- **Status:** not in VSX (checked 2026-11-02), GCVS, the ASAS-SN variable-star catalogue, the TESS eclipsing-binary catalogues of Prša et al. (2022) and Kostov et al. (2025), or Gaia DR3's eclipsing binaries.

Proposed VSX type EA, range 10.47 V (0.151) TESS.

The star

BD-13 1046 is a V = 10.47 star, probably an early-K dwarf from its TIC parameters, with a single-lined spectroscopic orbit in Gaia DR3.

Identifiers	BD-13 1046 = TIC 348773952 = TYC 5329-1207-1 = GSC 05329-01207 = 2MASS J05003818-1343082 = Gaia DR3 2987862104761137280
Position (J2000, Gaia DR3 at epoch 2000)	05 00 38.19, −13 43 08.4
Magnitudes	V 10.47 ± 0.02 (APASS DR9, 6 obs.); Tycho-2 VT 10.48; G 10.18; TESS T 9.60
Colours	B−V 0.90 (APASS DR9); J−K 0.59 (2MASS); BP−RP 1.14 (Gaia DR3)
TIC parameters	Teff 4912 K, R 0.74 R☉

Gaia DR3 SB1 orbit	P 5.07811 ± 0.00012 d, e 0.017 ± 0.007, K1 41.2 km/s (Gaia Collaboration, Arenou et al. 2023)
SIMBAD	SB*; associated with a ROSAT X-ray source

We have no spectrum of the star, so the VSX spectral-type field is left blank.

Discovery

The eclipses turned up in October 2026 in KESTREL, an automated search of TESS light curves for transiting planets. The signal was rejected as a planet because it is too deep and has a clear secondary eclipse. The search lists such stars for a separate check against VSX; this star was not in VSX, and the analysis below was done by hand.

TESS data and ephemeris

Sixteen primary eclipse times from 2020 to 2025 give P = 5.078277 ± 0.000001 d and Min I = BJD_TDB 2459990.01912 ± 0.00019, which is HJD (UTC) 2459990.01834.

Sector	Year	Product	Cadence	Used for
S98	2025	QLP	200 s	11 primary and 10 secondary eclipse times; depths
S32	2020	SPOC	120 s	5 primary and 5 secondary eclipse times; depths (corrected for crowding)
S5	2018	TESS-SPOC	1800 s	light curve only; no eclipse times

Each eclipse time comes from a trapezoid model whose shape is shared by all eclipses of that kind, with the time and depth free for each eclipse. A weighted straight-line fit to the 16 primary times over 367 cycles leaves an O−C scatter of 1.1 min (reduced χ^2 4.5). The quoted errors are scaled to match that scatter (by $\sqrt{4.5}$). The epoch in the VSX form is rounded to 2459990.018.

The eclipses

The primary eclipse is 0.151 mag deep in TESS and lasts 2.1% of the orbit; the secondary is 0.033 mag deep, exactly half an orbit later.

	Primary	Secondary
Depth, SPOC 120 s (S32)	12.97% = 0.151 mag (5 eclipses)	2.97% = 0.033 mag (5 eclipses)
Depth, QLP 200 s (S98)	10.62% = 0.122 mag (11 eclipses)	2.91% = 0.032 mag (10 eclipses)
Duration T14	2.56 ± 0.01 h	2.91 ± 0.04 h
Ingress and egress, each	1.03 ± 0.01 h	1.24 ± 0.04 h
Phase	0	0.5001

- **Depth used:** the SPOC value, which is corrected for light from other stars in the aperture. QLP's primaries are shallower and vary from 9.0% to 12.9% between eclipses, while its secondaries agree with SPOC; we have not traced the cause, so the VSX range uses SPOC.
- **Eclipse duration** for VSX: primary T14 / P = 2.56 h / 121.88 h = 2.1%.
- **Orbit shape:** the secondary falls at phase 0.5001, so $e \cos \omega \approx 0$. The fitted durations differ by 14%. Taken at face value, that would mean $e \sin \omega \approx 0.06$, more than Gaia's $e = 0.017 \pm 0.007$. The secondary is shallow, and its duration is tied to its ingress time in the fit, so its true uncertainty is probably larger than the quoted ±0.04 h. The eccentricity is left open, and it is not in the VSX remarks.
- **Between eclipses:** S32 shows a variation of about 1% (0.02 mag peak to peak) that does not repeat exactly from orbit to orbit, probably starspots.

Ground-based check: ASAS-SN V

ASAS-SN V photometry (2013 to 2018) is faint at the times of primary eclipse predicted by the TESS ephemeris, so the period holds back to 2013.

- **Data:** 331 V points from ASAS-SN's photometry database, source AP19370517 at 75.15943, −13.71894, HJD 2456311 to 2458448. Median V out of eclipse is 10.50 (rms 0.077), close to APASS's 10.47.
- **Primary eclipse:** the 7 points that fall inside primary eclipse, using the TESS ephemeris projected back about 720 cycles, average 0.13 mag fainter than the median. Only 0.7% of windows of the same width at other phases are as faint.
- **Limits:** the data are too sparse to measure the V depth or the secondary eclipse, so the VSX range keeps the TESS amplitude with the APASS V maximum.

The period error (0.000001 d) moves the predicted eclipse by about 1 minute over 720 cycles, which is much less than the 2.6-hour eclipse.

Which star varies

The eclipses are on BD-13 1046 itself, for two independent reasons.

1. **No neighbour is bright enough.** Every star within 1′ is at least 3.5 mag fainter in TESS, so it gives under 4% of the light. Such a star cannot cause a 13% dip even if it disappeared completely.
2. **The period is the star's own.** The eclipse period matches Gaia's spectroscopic orbit of BD-13 1046 to 1.4σ. A blended background binary would have no reason to share it.

Catalogue status

No variable-star or eclipsing-binary catalogue lists BD-13 1046; it is known only as a spectroscopic binary.

Catalogue	Result	Checked
VSX	not listed within 1′	2026-11-02
GCVS	not listed	2026-10-14
ASAS-SN variable stars	not listed	2026-10-14
TESS eclipsing binaries, Prša et al. (2022)	not listed; that catalogue covers 2-min data from S1 to S26, and this star's 2-min data start in S32	2026-10-14
TESS Ten Thousand Catalog, Kostov et al. (2025)	in none of its tables	2026-10-14
Gaia DR3 eclipsing binaries	not listed	2026-10-14
Gaia DR3 non-single stars	SB1 orbit (spectroscopic, not eclipsing)	2026-10-14
SIMBAD	type SB*; nine references, all surveys (RAVE, ROSAT, AllWISE, Gaia), none on eclipses	2026-10-14

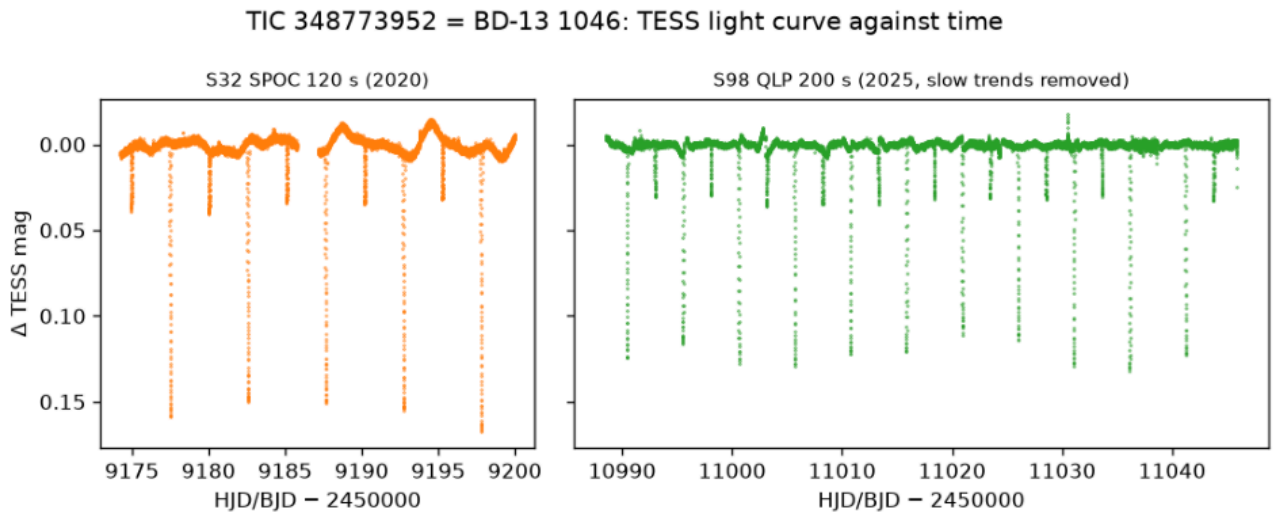
VSX entry

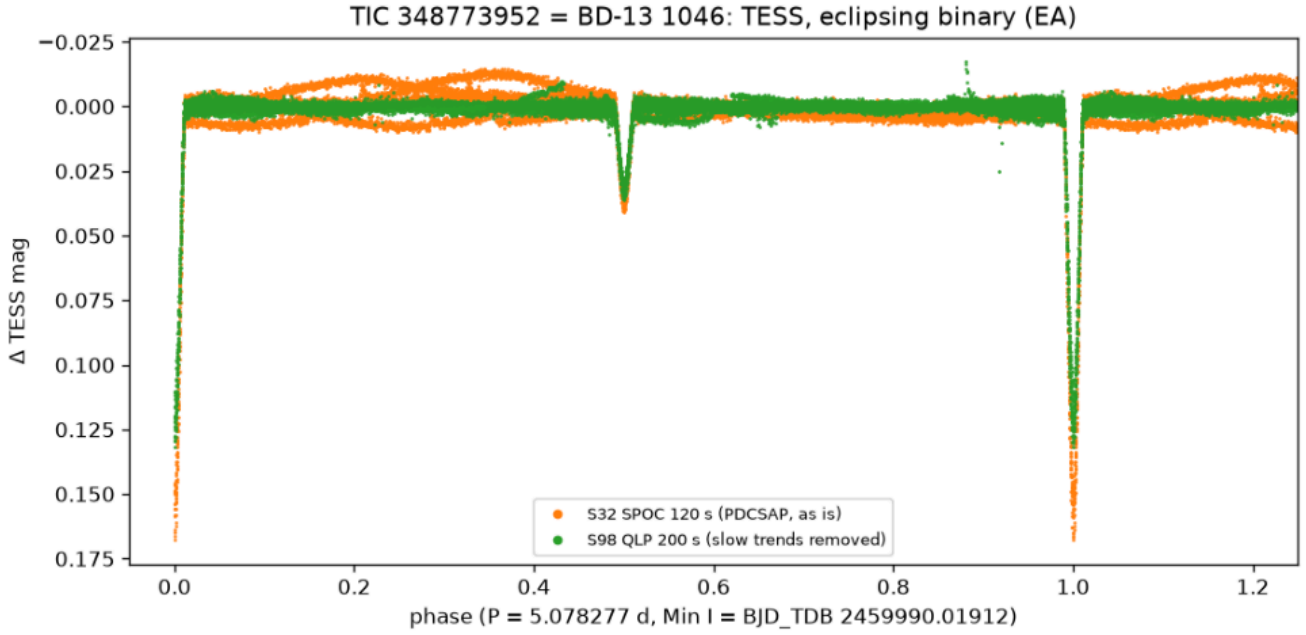
The submission gives BD-13 1046 as type EA, period 5.078277 d, range 10.47 V (0.151) TESS.

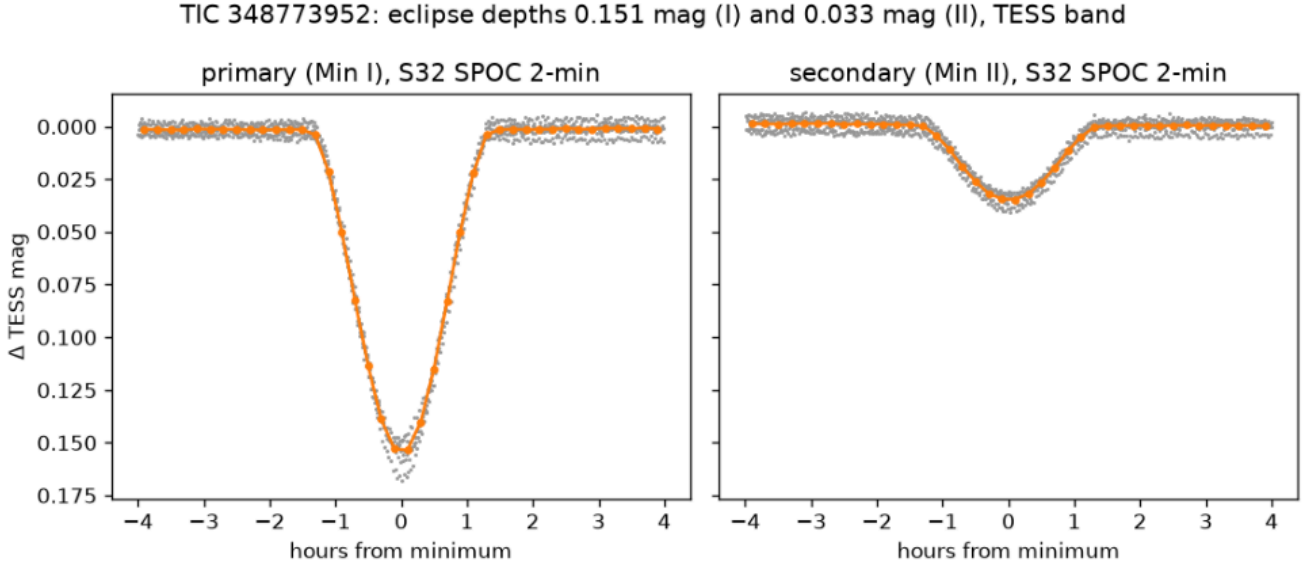
Field	Value
Name	BD-13 1046
Position (J2000)	05 00 38.19 −13 43 08.4 (Gaia DR3)
Other names	2MASS J05003818-1343082, TYC 5329-1207-1, GSC 05329-01207, TIC 348773952, Gaia DR3 2987862104761137280
Type	EA
Maximum	10.47 V (APASS DR9)
Minimum	0.151, as amplitude, passband TESS
Period	5.078277 d
Epoch (Min I)	HJD 2459990.018
Eclipse duration	2.1%
References	TESS (2014SPIE.9143E..20R); Gaia DR3 SB1 orbit (2023A&A...674A..34G)
Remarks	Spectroscopic binary (2023A&A...674A..34G). TESS amplitudes: 0.151 (I) and 0.033 (II); Min II at phase 0.500. Out-of-eclipse variation 0.02 TESS. B-V= 0.90 (APASS); J-K= 0.59 (2MASS).
Revision comment	Initial submission. Type, period, epoch and amplitudes from TESS data. Max. from APASS DR9. Gaia DR3 position.

The maximum is effectively the catalogue V, because the eclipses take only about 2% of the orbit.

Data, figures and references


TIC 348773952 = BD-13 1046: TESS light curve against time
S32 SPOC 120 s (2020)
S98 QLP 200 s (2025, slow trends removed)
Δ TESS mag
0.00
0.05
0.10
0.15
9175
9180
9185
9190
9195
9200
10990
11000
11010
11020
11030
11040
HJD/BJD − 2450000
HJD/BJD − 2450000


TIC 348773952 = BD-13 1046: TESS, eclipsing binary (EA)
Δ TESS mag
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
S32 SPOC 120 s (PDCSAP, as is)
S98 QLP 200 s (slow trends removed)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
phase (P = 5.078277 d, Min I = BJD_TDB 2459990.01912)


TIC 348773952: eclipse depths 0.151 mag (I) and 0.033 mag (II), TESS band
primary (Min I), S32 SPOC 2-min
secondary (Min II), S32 SPOC 2-min
Δ TESS mag
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
−4
−3
−2
−1
0
1
2
3
4
hours from minimum
hours from minimum

TESS light curve of BD-13 1046: S32 (2020, SPOC 120 s) and S98 (2025, QLP 200 s) against time; both folded at P = 5.078277 d; and close-ups of Min I and Min II from S32.

The files are in PlanetCandidates, vsx/TIC348773952.

File	Content
tic348773952_vsx_figure.png	The figure above (VSX supporting file 1)
tic348773952_tess_s32.txt	TESS S32 SPOC 2-min photometry: HJD (UTC), Δ TESS mag from the out-of-eclipse median, error (VSX supporting file 2)
vsx_submission.md	The VSX form values

References

- Gaia Collaboration, Arenou, F. et al. 2023, A&A, 674, A34 (2023A&A...674A..34G): Gaia DR3 orbits of non-single stars.
- Kochanek, C. S. et al. 2017, PASP, 129, 104502: the ASAS-SN light-curve server.
- Kostov, V. B. et al. 2025, ApJS, 279, 50 (2025ApJS..279...50K): the TESS Ten Thousand Catalog.
- Prša, A. et al. 2022, ApJS, 258, 16: TESS eclipsing binaries in sectors 1 to 26.
- Ricker, G. R. et al. 2014, Proc. SPIE, 9143, 914320 (2014SPIE.9143E..20R): TESS.
- Shappee, B. J. et al. 2014, ApJ, 788, 48: ASAS-SN.

APASS DR9 (VizieR II/336) and 2MASS photometry were taken from VizieR.